



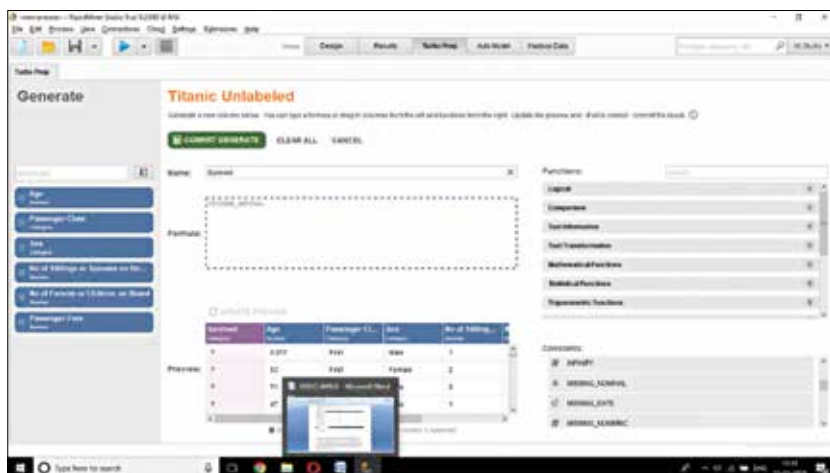
BMW's AI assistant for cars

A next-gen interface from BMW may let drivers interact with their cars like they're chatting with a human. <https://dgit.in/mar19-54>



Voice Typing comes to KaiOS phones

KaiOS, a popular new OS for feature phones like Jio Phone, will soon gain a major new features: YouTube, voice typing. <https://dgit.in/mar19-55>



Adding a new column

changes reflected. A new column will be added as shown above.

Now, we select the titanic table and click on Merge. We are provided with several options. We select merge with a second data set 'titanic unlabelled' and then select append as the merge type, update preview and finally commit. The two tables get merged with missing entries for the Survived column from the point where the titanic unlabelled merges with the titanic table. Now we need to click on model and it takes us to the auto model view. There are options for predicting, clustering and even detecting outliers in our data. For now, we want to see a predictive model. And finally we get to see our results.

It may take time to get the full analysis done. But the wait shall be worth it, indeed. We find tonnes of statistical information, especially on the accuracy and the run time of the models. An interesting thing to note here would be the predictions and the confidence attribute. Taking the first example, we understand that there is a 42.9 percent chance that this passenger had survived.

DECISION TREE – BUILDING THE MODEL

We also might want to have a decision tree model of our data, to have a wider perspective of the if's and else's that might arise. One thing to note is that complex models (like decision trees)

don't work well if we have missing values for the attribute of interest (survived, in our case). We might want to filter the data out for examples with no missing values for the survived column. To achieve this we use the retrieve operator and load our data. Next, we use the filter examples operator and set the survived to 'is not missing'. We can add multiple filters down the line and allow the behaviour to be a 'match all' or a 'match any' analogous to our Boolean AND and OR operators from digital logic.

To see the changes made in order to prepare our data, we can run the process in this state and check the statistics tab and look at the missing column. Data duplicates can also be found here. For instance, if we want to obtain a greater degree of statistics

for the 'age' attribute, we can click on statistics and we get to see a bar graph.

By clicking on Open Visualizations, we are welcomed by an even larger graphical histogram, which allows us to play with the attributes on either axis with any combinations of variables against the x and y axes. The usual formatting aspects for graphs like colours, titles, legends are all present and easy to use.

If we are done with this bit, we can go back to the design view and now search for the decision tree operator and drag it next to the filter examples operator. But hey, we get an error right away. It says 'input example set must have special attribute 'label''. We did talk of our attribute of interest, i.e., the survived attribute. But as of yet, the attribute is still being treated as a regular attribute. We need to change the role of this attribute. So, what we can do is go ahead and drag the Set Role operator and add it right after filter examples. For the parameters, we select the attribute name to be 'Survived' and the target role as 'label'. Now, any decision tree would be able to recognize this attribute and build a model likewise.

WHOA! Now isn't that a HUGE decision tree? This is because we had several of the attributes intact, attributes with almost all distinct values. However, if we insert the select attributes operator and select the attribute filter type as 'subset' and select the 'no correlation' attributes which we deem



Click on Statistics to see the statistics in a more detailed manner